



Increasing MTBF by Installing a Second Primary Fuel Filter on the 994K Wheel Loader

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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July 2023
1.10-230728-2

1.0 Introduction

A fleet of four 994K Wheel Loaders was introduced to a mine in Brazil in early 2020 in loaders feed mobile crushers and load off-road trucks at the mine. It is important to maintain feeding of the crushers to avoid interruption in the ore processing to support the crusher operation the loader operate in “load and carry” mode where the loader is constantly moving the material.

The mine was designed to be essentially “TRUCKLESS” so the mine does not have fixed fuel stations, causing that fuel is brought to the mine’s fuel distribution point and then transferred to other tanks and then to the loaders themselves, this results in poor fuel quality due to contamination and water condensation in tanks and ultimately in the loaders.

The loaders are equipped with 3516 engines with an average of 150 liter per hours of fuel consumption, in this fleet the Preventive Maintenances (PM’s) are performing every 250 hours and normally the fluids and filters would be replaced during the PM’s but because of the fuel quality and contamination issues the fuel filters reach saturation with contamination around ~125 hours.

As there is no by-pass in the systems, when a filter is clogged the machine is stopped due to low performance this “corrective stop” to replace the primary filter results in:

- a. lower MTBF,
- b. increased maintenance time and loss of production.

In this site, the client doesn't have fuel cleanliness system. It is very important and Sotreq also recommends to work on filtering the fuel before it is put into the machine.

2.0 Best Practice Description

Adding a second primary fuel filter in parallel in the fuel line is this Best Practice idea that was applied to the loader (and can be applied to other machines) to mitigate the poor fuel quality at site and the resulting losses in maintenance/performance and production.

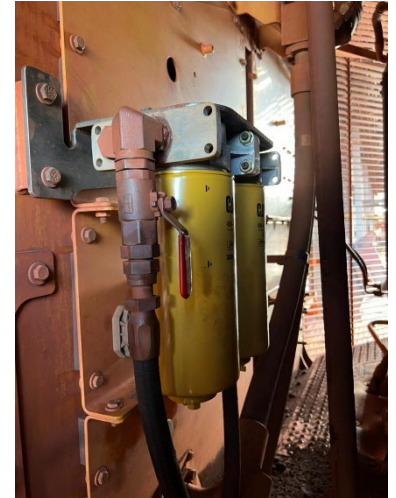
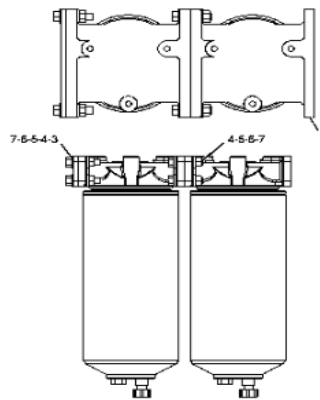
With the second primary fuel filter in place, fluid and filter changes can occur at the regular PM interval of 250 hours instead of 125 hours.

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Before (Only one primary fuel filter)



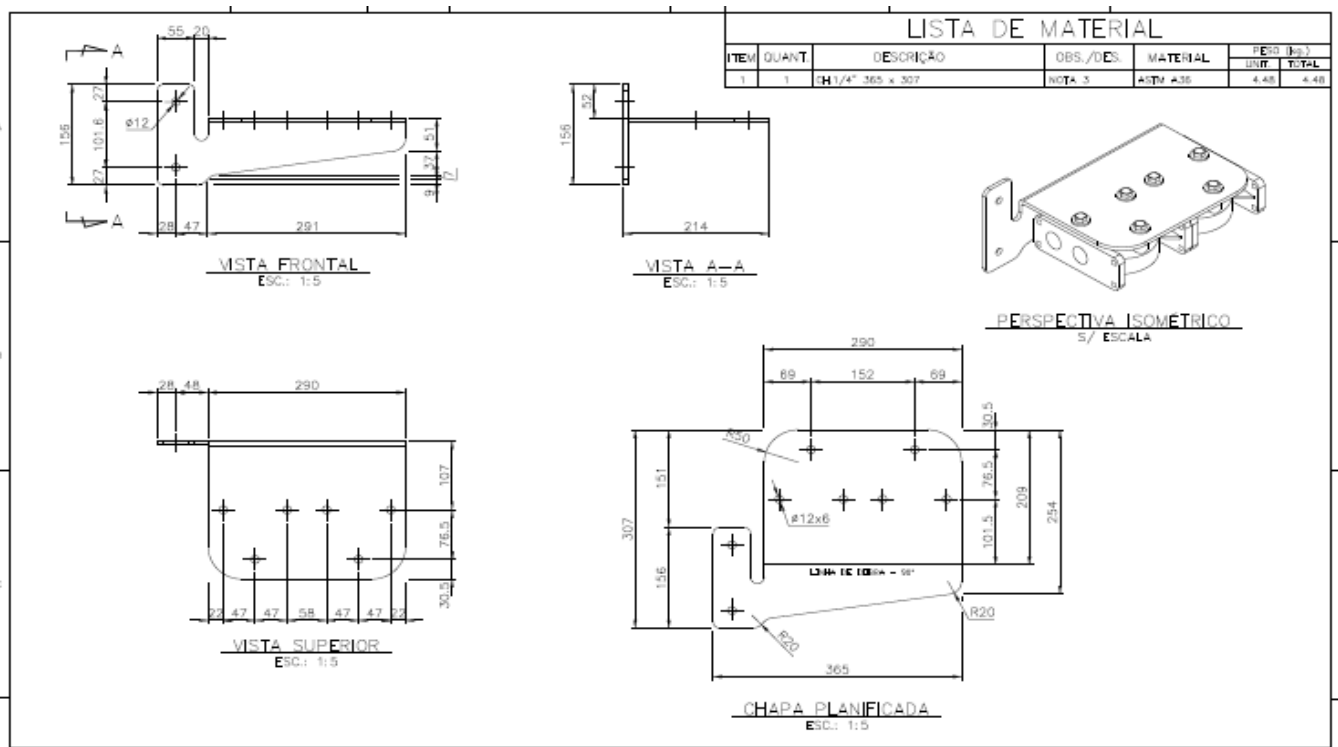
After (Two fuel filter in parallel)



3.0 Implementation Steps

After a Sotreq multidisciplinary team identified “poor fuel quality” as the main contributor to the need to change the fuel filter at 125 hours, a simple but effective solution was implemented in 3 steps:

- 1) **Procure a secondary primary filter (P/N: 438-5385)**
- 2) **Design and develop a support** to hold the two primary filter solution, this support bracket was developed by Sotreq is shown below.



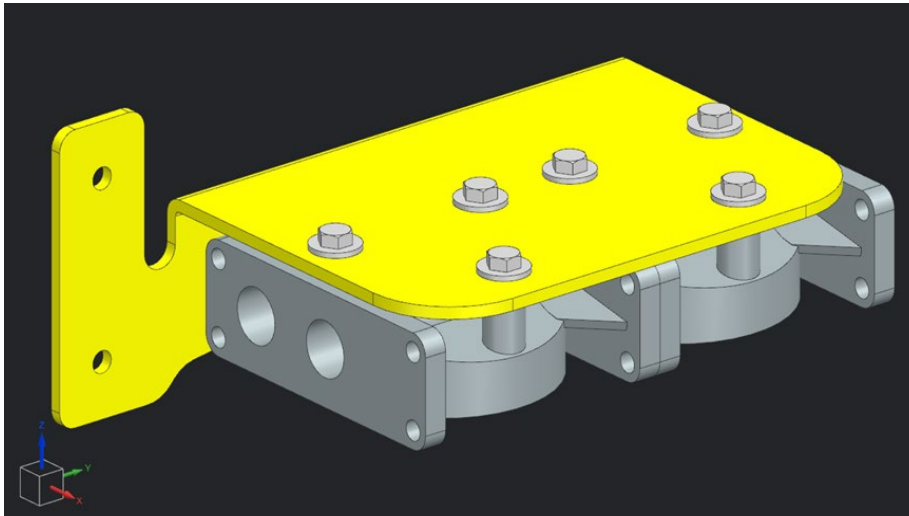
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Project new support



Support after install

- 3) **Test and Monitor the Solution:** With the implemented two primary filter design, the fluid and filter changes are again being done at the 250 hours PM's instead of at ~120 hours corrective maintenance stop.

Important: No bracket or chassis issues have been encountered after

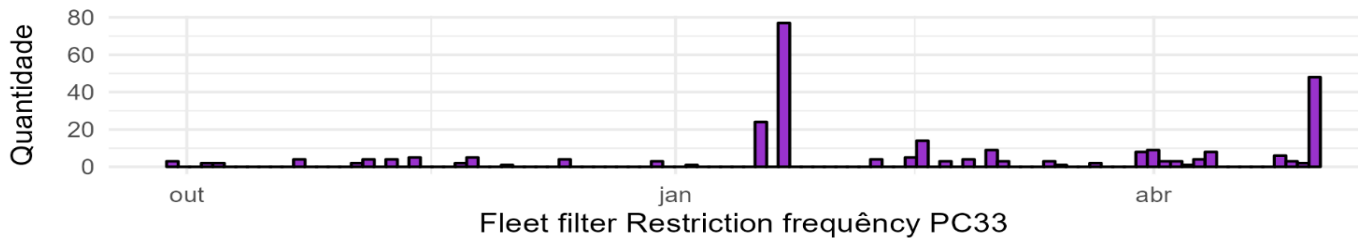
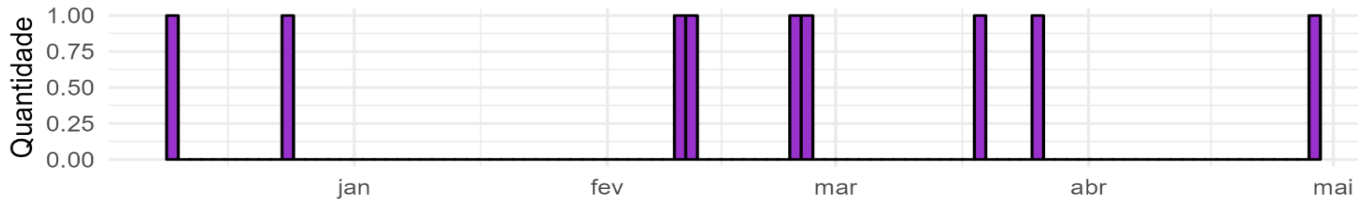
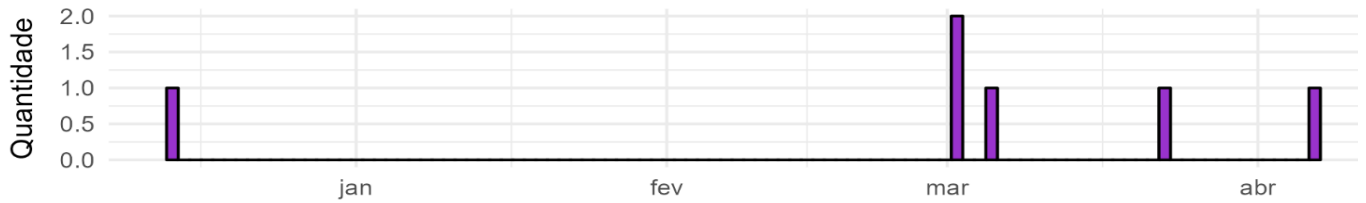
Equipment	TAG	Operating hours after install modification
MRK00332	PC30	5280 hours
MRK00333	PC31	3345 hours
MM900315	PC32	Waiting for application
MM900320	PC33	485 hours

In this chart below it is possible to see the events number of “**Fleet Fuel Filter Restriction**” and “**Low Fuel Pressure**” reduced significantly after modifying another filter in the fuel line.

Note: PC30 have no alarms during this period, the installation on it machine occurred before all other machines in fleet in 2020.

Fleet Fuel Filter Restriction + Low Fuel Pressure during evaluation

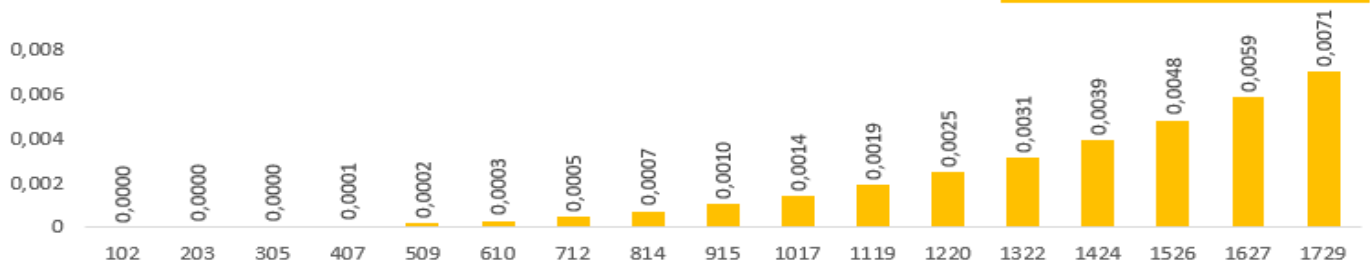
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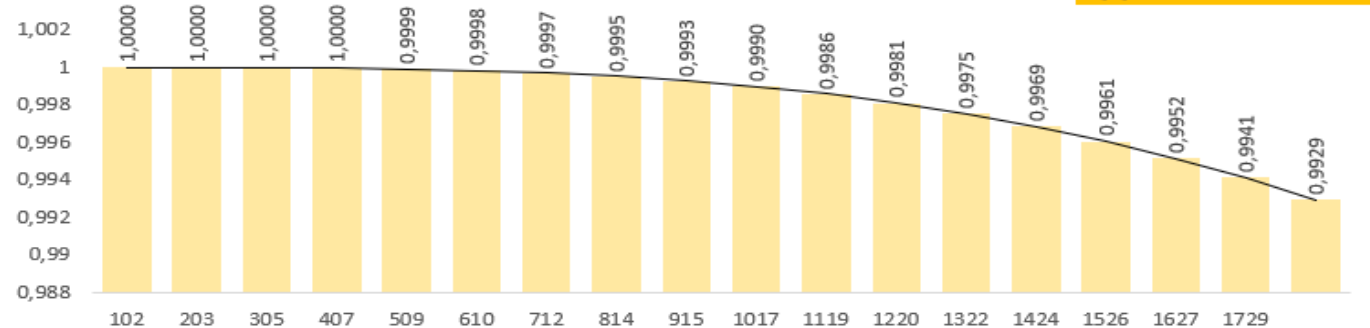
The next graphic is presented reliability and failure probability “**For Fuel Filter**” in PC32, the unique machine without second filter until this moment.

Reliability behavior to **PC32** (Without second filter)

F(t) - Probabilidade Falha



R(t) - Confiabilidade



The next chart shows the reliability and failure probability for “**For Fuel Filter**” in PC30, after installing the second filter from the start of operation. in this behavior is clear the result with second filter is better them PC33.

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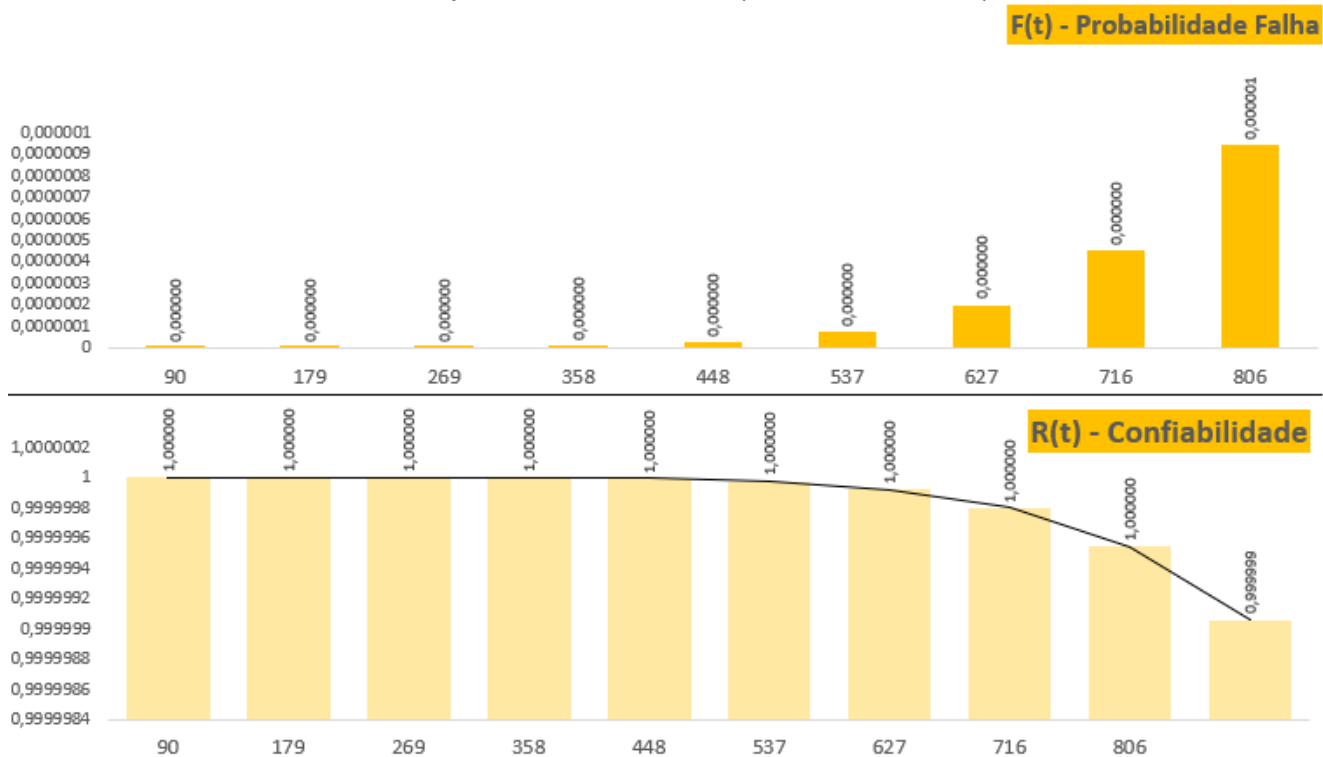
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Reliability behavior to **PC30** (With second filter)



Note: The reliability and failure probability is only for fuel filter group at 994 fleet. Additionally, Sotreq has been working with the customer to better understand the machine maintenance / performance issues related to the poor fuel quality and consider additional steps for control contamination at the site.

As a result, in addition to the two primary filter system on the loader, the customer is in the process of implementing a fuel station where contamination can be better controlled as well as other contamination control procedures throughout the fueling process.

4.0 Benefits

Benefits:

- Less Maintenance / Increase MTBS Performance
- Increase Productivity / Production

Less corrective maintenance to replace the primary fuel filter, consequent, increasing the value of MTBS and decreasing the accident risk of the technician exposure to field work. Furthermore, increasing of equipment overall effectiveness.

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5.0 Resources Required

Computer with drawing software, metal plate, filter base.
Labor (Fabrication/Assembly): 08 hours – 3000,00 BRL

Parts need:

PN	Qty	Description	Price
2256948	4	SEAL-O-RING	R\$ 57,63
8T4182	4	BOLT	R\$ 6,35
8T4121	8	WASHER-HARD	R\$ 3,01
8T4133	4	NUT	R\$ 2,11
3211326	1	BASE AS FILTER	R\$ 6.489,87

The total cost is the cost of the parts, plus the cost of the labor:

$$6558,97 \text{ BRL} + 3000,00 \text{ BRL} = \mathbf{9558,97 \text{ BRL (~2000 USD)}}$$

6.0 Supporting Attachments / References

- Drawing attached

7.0 Related Best Practices

N/A

8.0 Acknowledgements

SOTREQ would like to thank the immense commitment of Engineering team, always willing to deliver the best fleet performance possible by developing strategic solutions.

We would like to mention the close collaboration of **Doug Friedrich** of Caterpillar with our company, thanks Doug!

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